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Inverse Kinematics of 7-DOF Redundant Manipulators with Arbitrary Offsets Based on Augmented Jacobian

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***Abstract*—To solve the inverse kinematics of 7-DOF redundant manipulators with arbitrary offsets. A numerical method that is based on augmented Jacobian is proposed. Firstly, the end-effector Jacobian is given to reflect the relationship between joint angle rates and end-effector rates. To solve the redundancy of 7-DOF manipulator, an arm angle parameter is defined to parameterize the redundancy of manipulator. From the geometric features of the manipulator, the Jacobian matrix of arm angle is built to solve the redundancy of the manipulator. Furthermore, singularities of Jacobian are discussed mainly by analyzing the rank of Jacobian. In the end, the method is verified with Yaskawa VA1400II manipulator. The results of simulations and experiment strongly prove the validity and efficiency of this method.**

Keywords—inverse kinematics, arbitrary offsets, arm angle, augmented Jacobian, singularity analysis

I. INTRODUCTION

The traditional 6-DOF manipulator is used in many industrial places to accomplish different tasks. Since the manipulator has 6 degrees of freedom, it is just enough to reach a position and orientation in Cartesian space. For some tasks such as obstacle-avoidance[1] and singularity avoidance[2] that demand for different structure of manipulator while reaching the target position at the same time. Thus, the 6-DOF manipulator is not suitable, and the 7-DOF manipulator is proposed. The extra DOF of manipulator greatly increases flexibility and manipulability of the manipulator. With the redundant joint, such manipulators can accomplish those specific tasks . However, the inverse kinematics of 7-DOF manipulators is more than complicated. Many studies have discussed inverse kinematics of 7-DOF manipulators with different solutions.

To solve the inverse kinematics of 7-DOF manipulators with arbitrary offsets, Kircanski [3] proposed a numerical method that is based on Jacobian matrix of end-effector. However, since the Jacobian $\in R^{6*7}$, the method relies on pseudoinverse , and the computation of this method is time-consuming. The redundancy of 7-DOF manipulators is lost because the method can only obtain one feasible solution for inverse kinematics. To solve the problem of computation time, Lian[4] proposed a parameter to accelerate the iteration of pseudoinverse matrix, but the redundancy of remains to be solved. Shimizu and Kakuya[5] solved analytical inverse kinematic solution of S-R-S 7-DOF manipulators, which consists of spherical shoulder, spherical wrist and elbow, by fixed-arm-angle method. The arm angle is defined between arm plane which contains shoulder, elbow and wrist and reference plane. Wu[6] and Yu[7] proposed a method to find analytical inverse kinematic solution of manipulators similar to S-R-S type with offset wrist. Furthermore, Luo[8] found the analytical inverse kinematic solution of manipulators with offset shoulder and offset wrist. By applying the method, analytical solution for this type of arm still can be found with fixed-arm-angle. But the analytical solution is only applicable to the manipulators similar to S-R-S type. Fig. 1 shows the generic structure of 7-DOF manipulators with offset shoulder, offset elbow, and elbow wrist. Since the structure does not meet Pieper standard, it is impossible to find a closed-form solution for inverse kinematics. YAN[9] proposed an optimization method for inverse kinematics of 7-DOF manipulator with arbitrary offsets. By fixing one joint, the inverse kinematics of manipulator can be treated as inverse kinematics of 6-DOF manipulator. However, this method loses the redundancy of 7-DOF manipulators, and the analysis is too complicated.

To solve the inverse kinematics of 7-DOF manipulators with arbitrary offsets as well as utilize the redundancy of manipulators, the paper proposes a method based on augmented Jacobian. Learning from numerical method, we proposed the End-effector Jacobian. To utilize the redundancy of manipulator, the arm angle is defined and the Jacobian matrix of arm angle is built. In section II, the Denavit-Hartenberg parameters and the model of Yaskawa V11400II are shown and the forward kinematics is computed. Section III shows the method to solve inverse kinematics based on augmented Jacobian. Section IV analyzes the singularity by analyzing the rank of Jacobian and provides simulation and experimental result of inverse kinematic solution. Section V is the concluding remarks.

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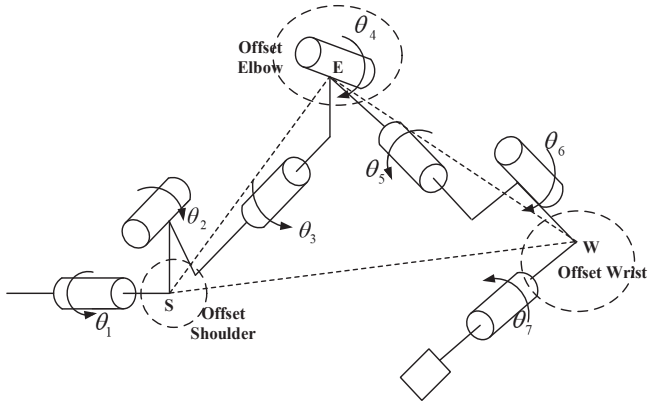

Fig. 1. Generic structure of 7-DOF manipulator

II. FORWARD KINEMATICS

A. Model of VA1400II

To analyze the kinematics of manipulator efficiently, the model of VA1400II is shown in Fig. 2, where the shoulder, elbow and wrist represent the origins of l, 4 and 7 respectively. Unlike the S-R-S structure of manipulators, the manipulator has offsets at shoulder, elbow and wrist, and the offsets are a_1 , a_4 and a_6 .The D-H parameters are shown in Table 1..

B. Forward Kinematics of VA1400II

With D-H parameters, the forward kinematics can be solved by homogenous transformation matrix[10]which is given by:

$$ {}^{i-1}T_i = \begin{pmatrix} {}^{i-1}R_i & {}^{i-1}P_i \\ 0 & 1 \end{pmatrix} \tag{1} $$

$$ {}^{i-1}R_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i & 0 \\ \sin\theta\cdot\cos\alpha_{i-1} & \cos\theta_i\cdot\cos\alpha_{i-1} & -\sin\alpha_{i-1} \\ \sin\theta\cdot\sin\alpha_{i-1} & \cos\theta_i\cdot\sin\alpha_{i-1} & \cos\alpha_{i-1} \end{bmatrix} \tag{2} $$

$$ {}^{i-1}P_i = \begin{bmatrix} a_{i-1} \\ -d_i\cdot\sin\alpha_{i-1} \\ d_i\cdot\cos\alpha_{i-1} \end{bmatrix} \tag{3} $$

where a_i , d_i and α_{i-1} are D-H parameters, ${}^{i-1}R_i$ and ${}^{i-1}P_i$ give the transformation from the (i-1) axis to the i axis From (4), we can obtain ${}^0R_7$ and ${}^0P_7$ which can be used as known parameters to solve inverse kinematic problem.

$$ {}^0T_7 = {}^0T_1\cdot{}^1T_2\ldots\cdot{}^6T_7 \tag{4} $$

Table 1 . D-H parameters for VA1400II manipulator

i	α_{i-1}	a_{i-1}	d_i	θ	
1	0°	0	0	θ_1	a_1=15cm
2	-90°	a_1	0	θ_2	a_4= -20cm
3	+90°	0	d_3	θ_3	a_6= -3cm
4	-90°	0	0	θ_4	d_3 =61.4cm
5	+90°	a_4	d_5	θ_5	d_5 =64cm
6	-90°	0	0	θ_6	
7	+90°	a_6	0	θ_7	

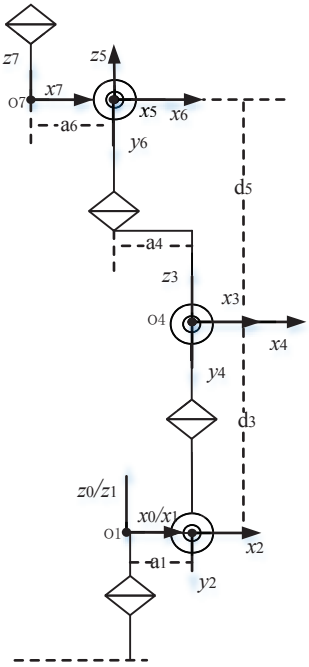

Fig. 2. Model of VA1400II with offsets at shoulder, elbow and wrist

III. INVERSE KINEMATICS OF VA1400II

With a given end-effector position and arm angle ψ , there will generally exist one group of inverse solution. To obtain the inverse solutions, it is important to find the End-effector Jacobian J^{ee} and the Jacobian matrix of arm angle J^{ψ} . The augmented Jacobian matrix is defined by:

$$ J^A = \begin{pmatrix} J^{ee} \\ J^{\psi} \end{pmatrix} \tag{5} $$

and $\dot{X} = J^A\dot{\theta}$, $\dot{X}$ represents the rate of change of the end-effector and the arm angle ψ , $\dot{\theta}$ represent the joint angle rates of manipulator. With the given augmented matrix J^A , the inverse kinematics of VA1400II can be solved by iteration algorithm.

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A. End-effector Jacobian

The end-effector Jacobian matrix gives end-effector rates as functions of joint angles. Generally, the end-effector Jacobian J^{ee} is equal to ${}^0J_7$ and it can be derived from transformation matrix ${}^{i-1}T_i$. The end-effector Jacobian [11] is shown as:

$$ {}^0J_7 = \begin{pmatrix} {}^0\hat{z}_1 & \dots & {}^0\hat{z}_7 \\ {}^0\hat{z}_1 \times {}^0P_{7,1} & \dots & {}^0\hat{z}_7 \times {}^0P_{7,7} \end{pmatrix} \tag{6} $$

where ${}^0P_{i,j} = {}^0P_i - {}^0P_j$ and ${}^0\hat{z}_i = {}^0R_i e_3$, $e_3 = (0,0,1)^T$ .From (2) and (3) , the Jacobian matrix of end-effector can be computed. However, it is tedious and time-consuming to find such Jacobian matrix directly.

To simplify the computation of Jacobian, Fijany[11] proposed a method that the Jacobian ${}^bJ_a$ gives the rate of change of frame "a" represented in frame "b", and ${}^bJ_a$ is related to ${}^dJ_c$ which gives the rate of change of frame "c" represented in frame "d". The relationship between ${}^bJ_a$ and ${}^dJ_c$ can be found as:

$$ {}^bJ_a = \begin{pmatrix} {}^bR_d & 0 \\ 0 & {}^bR_d \end{pmatrix} \cdot \begin{pmatrix} I & 0 \\ {}^d\tilde{P}_{a,c} & I \end{pmatrix} \cdot {}^dJ_c \tag{7} $$

where, for a 3-vector $x = [x1, x2, x3]^T$, $\tilde{x}$ denotes 3*3 skew symmetric matrix defined by:

$$ \tilde{x} = \begin{bmatrix} 0 & x3 & -x2 \\ -x3 & 0 & x1 \\ x2 & -x1 & 0 \end{bmatrix} \tag{8} $$

${}^bR_d \in R^{3\times3}$ is a rotation that gives the orientation of frame "d" represented in frame "b". In the paper, we assume frame "a", "b", "c" and "d" represent frames 7, 0, 3 and 3 respectively. The end-effector Jacobian could be given by

$$ {}^0J_7 = \begin{pmatrix} {}^0R_3 & 0 \\ 0 & {}^0R_3 \end{pmatrix} \cdot \begin{pmatrix} I & 0 \\ {}^3\tilde{P}_{7.3} & I \end{pmatrix} \cdot {}^3J_3 \tag{9} $$

where ${}^3P_{7,3} = {}^0R_3^T({}^0P_{7,0} - {}^0P_{3,0})$, resulting in

$$ {}^3P_{7,3} = \begin{bmatrix} -s_4(a_6 s_6 - d_5) + c_4(a_4 + c_5 c_6 a_6) \\ s_5 c_6 a_6 \\ -c_4(a_6 s_6 - d_5) - s_4(a_4 + c_5 c_6 a_6) \end{bmatrix} \tag{10} $$

and ${}^0R_3$ is given by ${}^0R_3 = {}^0R_1\,{}^1R_2\,{}^2R_3$

$$ {}^0R_3 = \begin{bmatrix} c_1c_2c_3 - s_1s_3 & -c_1c_2s_3 - s_1c_3 & c_1s_2 \\ c_1s_3 + s_1c_2c_3 & c_1c_3 - s_1c_2s_3 & s_1s_2 \\ -s_2c_3 & s_2s_3 & c_2 \end{bmatrix} \tag{11} $$

Note that $c_i = \cos(\theta_i)$, $s_i = \sin(\theta_i)$ and θ_i represents the angle of joint "i". From (7) , it can be found that $rank({}^3J_3) = rank(J^{ee})$ which means that we can analyze the singularity of end-effector Jacobian from ${}^3J_3$ .The symbolic expression for ${}^3J_3$ is given by

$$ {}^3J_3 = \begin{bmatrix} -s_2c_3 & s_3 & 0 & 0 & s_4 & -c_4s_5 & c_4c_5s_6 + s_4c_6 \\ s_2s_3 & c_3 & 0 & 1 & 0 & c_5 & s_5s_6 \\ c_2 & 0 & 1 & 0 & c_4 & s_4s_5 & c_4c_6 - s_4c_5s_6 \\ d_3s_2s_3 + a_1 s_3 & d_3c_3 & 0 & 0 & 0 & a_4s_4c_5 - d_5c_4c_5 & c_4s_5a_6 + a_4s_4s_5s_6 - d_5c_4s_5s_6 \\ (d_3s_2 + a_1)c_3 & -d_3s_3 & 0 & 0 & -a_4 & -s_5d_5 & c_5(s_6 d_5 - a_6) - a_4 c_6 \\ 0 & 0 & 0 & 0 & 0 & a_4c_4c_5 + d_5s_4c_5 & s_5[a_4c_4s_6 + s_4(s_6 d_5 - a_6)] \end{bmatrix} \tag{12} $$

The end-effector Jacobian J^{ee} can be computed from (7) and (10).

B. Arm Angle Jacobian J^{ψ} and Augmented Jacobian

As is shown in Fig. 3, where $S = O_1$, $E = O_4$,and $W = O_7$ denote the origins of frames 1, 4, and 7 attached to the shoulder elbow and wrist respectively[5], the arm angle ψ is defined by the angle from the reference plane containing the unit vector V and the shoulder-wrist line SW to the

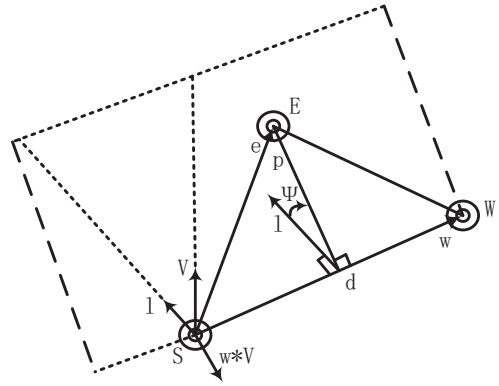

Fig. 3. Definition of the arm angle ψ

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shoulder-elbow-wrist plane SEW. To find the relationship between arm angle ψ and joint angles, let vector $w = W - S$, $e = E - S$, and $V = (0,0,1)^T$. The projection of e onto w is given by $d = \overline{w}(\overline{w}e)$, $\overline{w} = w / \|w\|$. The minimum distance from the line SW to the point E is the vector $p = e - d = (I - \overline{w}\overline{w}^T)e$. The reference plane contains both w and V, and the vector in the reference plane that is orthogonal to w is given by $l = (w \times v) \times w$. We define unit vectors $\overline{p} = p / \|p\|$ and $\overline{l} = l / \|l\|$. Notes that e, w, d, p and l can be computed from (2). The arm angle ψ is the angle from l to p, this construction results in

$$C_\psi = \overline{l}^T \overline{p}, \qquad S_\psi = \overline{w}^T (\overline{l} \times \overline{p}) \tag{13}$$

where $C_\psi = \cos\psi$ and $S_\psi = \sin\psi$. From (13), arm angle is derived from

$$\psi = a\tan 2(\overline{w}^T (\overline{l} \times \overline{p}), \overline{l}^T \overline{p}) \tag{14}$$

In order to proceed further, it is necessary to define the Jacobian matrices E and W that relate joint rates to $\dot{e}$ and $\dot{w}$. $\dot{e}$ is the linear velocity of the elbow point E, and $\dot{w}$ is the linear velocity of the elbow point W. From Seraji[13], Jacobian matrices E and W are given by

$$E = \begin{bmatrix} {}^0\hat{z}_1 \times {}^0P_{4,1} & {}^0\hat{z}_2 \times {}^0P_{4,2} & {}^0\hat{z}_3 \times {}^0P_{4,3} & 0 & \dots & 0 \end{bmatrix} \tag{15}$$

$$W = \begin{bmatrix} {}^0\hat{z}_1 \times {}^0P_{7,1} & \dots & {}^0\hat{z}_6 \times {}^0P_{7,6} & 0 \end{bmatrix} \tag{16}$$

Because the arm angle ψ is given by the angle from l to p, it is natural that the arm angle Jacobian should depend on l and p, this can result in

$$\begin{aligned} J^\psi &= \frac{1}{\|p\|}(\overline{w} \times \overline{p})^T \dot{p} - \frac{1}{\|l\|}(\overline{w} \times \overline{l})^T \dot{l} \\ &= \frac{1}{\|p\|}(\overline{w} \times \overline{p})^T \{E - \frac{\overline{w}^T e}{\|w\|} \mathrm{W}\} + \frac{V^T w}{\|l\|}(\overline{w} \times \overline{l})^T W \end{aligned} \tag{17}$$

From (6) and (17), augmented Jacobian can be obtained and the inverse kinematics of VA1400II can be solved.

C. Iteration Algorithm Based on Augmented Jacobian

With the given augmented Jacobian J^A, the inverse kinematics of VA1400II can be solved by iteration of Jacobian. The main process of iteration algorithm[4] is shown in Table 2, where the Max iterations is the maximum times of iteration and the threshold is the allowable error defined by the difference between target position and actual position of manipulator. Usually, the *Max* iterations and threshold are set according to the demand of accuracy and speed of algorithm.

Table 2. Iteration algorithm of augmented Jacobian

Iteration algorithm

Input: X_t, ψ_t :Target Position and Arm Angle

Output: θ :Updated Joint Angles

1. Set θ_0 Through Start Position and ψ_s
2. **for** i from 1 to Max iterations **do**
3. Compute Augmented Jacobian
4. $\Delta\theta_i = J^{A^{-1}}[X_t - X_s, \psi_t - \psi_s]^T$
5. $\theta_i = \theta_{i-1} + \Delta\theta_i$
6. Compute updated X_i and ψ_i from updated θ_i
7. Error $e_i = \|[X_t - X_i, \psi_t - \psi_i\|$
8. **If** e_i <threshold e **then**
9. **Return** θ_i
10. **If i**> Max iterations
11. **Return "no solution"**
12. **end**
13. **end**

IV. Singulatities Analysis and Experiment

From section III, the inverse kinematics of VA1400II can be obtained in most cases. However, if the initial joint angles θ_0 are set improperly, the inverse matrix of J^A may not exist since $\mathrm{rank}(J^A) < 7$, and the augmented Jacobian is at a singularity. So it is necessary to discuss singularities of augmented Jacobian.

A. Singularities of Augmented Jacobian

The singularities of augmented Jacobian are of two types: the "kinematic singularities"[12] of the end-effector and additional "algorithmic singularities" [13]caused by arm angle ψ. A measure of nearness of redundant manipulator to kinematic singularity is defined by

$$m(\theta) = \left[\det(J^{ee}(\theta) J^{eeT}(\theta))\right]^{\frac{1}{2}} \tag{18}$$

If the manipulator is not at a kinematic singularity, $m(\theta) \neq 0$ and $\mathrm{rank}(J^{ee}) = 6$. When the parameter $m(\theta)$ is close to zero, the manipulator is nearly at a kinematic singularity[13]. This parameter can be used to discuss whether manipulator is about at a kinematic singularity. If $\mathrm{rank}(J^{ee}) < 6$, the manipulator is at kinematic singularity and there exists at least one direction along which the end-effector cannot move and the number of directions is

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$6-\mathrm{rank}(J^{ee})$. For example, it can be found from(10) that if all joint angles are equal to zero, $\mathrm{rank}(J^{ee})=5$ and the manipulator is at a kinematic singularity.

If $\mathrm{rank}(J^{ee})=6$, there exists a unit vector $n \in R^7$ that $J^{ee}n=0$. It can be proved that if and only if

$$J^{\psi}=kn^T \quad (k \neq 0) \tag{19}$$

the manipulator is not at algorithmic singularity[14] since J^{ψ} is independent of J^{ee} and $\mathrm{rank}(J^A)=7$. The vector n gives the direction along which the manipulator can move without causing end-effector motion[15]. From (12) and (17), kinematic and algorithmic singularities of VA1400II can be proposed. However, since the augmented Jacobian is extremely complicated, the analysis is more than complicated and remains to be complete. Generally, to avoid singularities, joint angles can be set that ensure $\mathrm{rank}(J^A)=7$.

B. Simulation and Experiment

To verify the algorithm, points p1,p2 and p3 are arbitrary set in task place. Assuming that $p1=[1.1993 \quad 0 \quad 0.1018]^T$, $p2=[1.0387 \quad 0.5997 \quad 0.1018]^T$, $p3=[0.5997 \quad 1.0387 \quad 0.1018]^T$. Since p1, p2, and p3 are not collinear, there exists a arc through three points[14]. This arc is divided into fifty points considered as target points of manipulator on average. The initial joint angles are set by (0^0, 45^0, 0^0, 90^0, 30^0, 43^0, -28^0) and the arm angle ψ set by zero. From section III, the inverse kinematics of VA1400II can be computed. As is shown in Fig. 4, the joint angles of manipulator vary smoothly, there does not exist mutation of joint angles, it proves that the method can ensure the 7-DOF manipulator move smoothly. Fig. 5 shows the end-effector positions that is consistent with the given target position. This proves that the method meets the demand of accuracy for inverse kinematics.

. If the arm angle ψ is given differently, there will exist different solutions for inverse kinematics of VA1400II without changing the end-effector position. This is called "Null Space Motion".

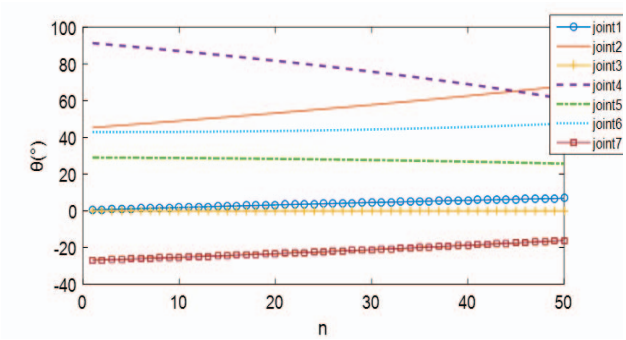

Fig. 4. Joint angles of VA1400II

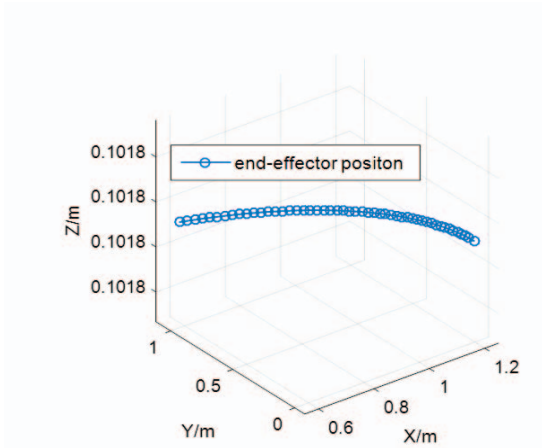

Fig.5. End-effector position of VA1400II

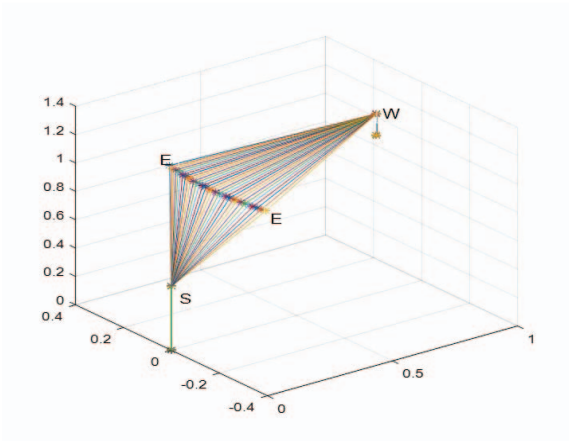

Fig. 6. S-E-W structure of VA1400II

Fig. 7. Joint angles related to "Null Space Motion"

Fig. 6 shows the different manipulator structure of S-E-W plane which contains shoulder, elbow and wrist with different arm angle ψ ($\psi \in (-30^0,+30^0)$) while the end-effector position remains the same. And Fig. 7 shows joint angles of the manipulator with different arm angle while the end-effector position is set from(18). The "Null Space Motion" can be used accomplish some tasks such as welding and obstacle-avoidance that demand for different structure of arm while the end-effector position remains the same.

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Fig. 8. Model of VA1400II manipulator

Table 3. Joint Angles of VA1400II

ψ^0	-20	-10	0	10	20
θ_1	39.34	23.88	0	-23.88	-39.31
θ_2	5.88	1.67	0	1.67	5.88
θ_3	-52.54	-29.97	0	29.97	52.51
θ_4	86.07	88.52	90.01	88.52	86.06
θ_5	4.66	0.84	0	-0.84	-4.66
θ_6	90.35	90.03	89.99	90.03	90.35
θ_7	-13.03	-6.07	0	6.07	13.03

To verify the validity of this method for inverse kinematics of real manipulators, the experiment is conducted in the laboratory with the VA1400II that is shown in Fig. 8. The target position is set by:

$$T = \begin{bmatrix} 1 & 0 & 0 & 0.82 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0.814 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{18}$$

The initial joint angles are set by (0^0, 45^0, 0^0, 90^0, 30^0, 43^0, -28^0). With the different arm angle ψ , different solutions for inverse kinematics of manipulator can be obtained. The joint angles of joint angles are shown in Table 3.

V. Conclusion

In this paper, a method based on augmented Jacobian is derived to solve inverse kinematics of 7-DOF manipulators with arbitrary offsets. Compared to iteration algorithm with pseudoinverse , this method is more fast if singularities are avoided. With different arm angle ψ the redundancy of 7-DOF manipulators can be utilized. However, The problem of singularity avoidance remains to be accomplished because singularities of Jacobian are related to the structure of different manipulators. For example, if all offsets of manipulator are all equal to zero in Table 1, it is convenient to find the singularity of end-effector Jacobian , but for a generic manipulator, the discussion is more complicated. This method is verified by simulation and experiment and is proved to be applicable and efficiency.

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